

Sec: XI_STU_IIT_IC
Time: 1.5 Hrs.

FTM-1_WARM-UP

Date: 03-06-2024
Max. Marks: 180

JEE MAIN 2022 Model PHYSICS

Section	Question Type	+Ve Marks	-Ve Marks	No. of Qs.	Total marks
Sec – I (Q.N : 1 – 10)	Questions with Single Correct Choice	4	–1	10	40
Sec-II (Q.N : 11 – 15)	Questions with Numerical Answer Type(+ / –Number)	4	–1	5	20
Total				15	60

CHEMISTRY

Section	Question Type	+Ve Marks	-Ve Marks	No. of Qs.	Total marks
Sec – I (Q.N : 16 – 25)	Questions with Single Correct Choice	4	–1	10	40
Sec-II (Q.N : 26 – 30)	Questions with Numerical Answer Type(+ / –Number)	4	–1	5	20
Total				15	60

MATHEMATICS

Section	Question Type	+Ve Marks	-Ve Marks	No. of Qs.	Total marks
Sec – I (Q.N : 31 – 40)	Questions with Single Correct Choice	4	–1	10	40
Sec-II (Q.N : 41 – 45)	Questions with Numerical Answer Type(+ / –Number)	4	–1	5	20
Total				15	60

Exam Syllabus

PHYSICS	50%: MATHEMATICAL PHYSICS : Application of differentiation in physics, Maxima & Minima Integration and its application in physics. 50%: Unit and Dimension : Introduction of physics, physical quantities and physical laws Fundamental and derived quantities and ways to write dimension, Unit and Dimension Uses of dimensional analysis (Principle of homogeneity)
CHEMISTRY	50% : Nature of Matter, Properties of matter and their measurement SI unit, Mass weight, Volume temperature, Need for standard reference, Scientific notation, Precision and accuracy, Significant figure and calculation involving significant figure. Law of chemical combination : (i) Law of conservation of mass, (ii) Law of constant combination, (iii) Law of multiple proportion, (v) Gay Lussac's law, (vi) Avagadro law, (vii) Dalton's atomic theory Atomic mass 50% : Mole concept: Calculation based on mole concept, concept of gram atom, gram molecule Percentage by mass (w/w), percentage by volume (v/v)
MATHEMATICS	50%: SEQUENCE AND SERIES Introduction of Sequence and Series, nth term of AP. Properties of A.P. ,Sum of n term of an AP, Arithmetic Mean. 50%: nth term of G.P.,sum of n term of GP. Properties of G.P., Geometric mean, A.G.P.

PHYSICS**Max. Marks: 60**

- When a wave traverses a medium, the displacement of a particle located at 'x' at a time 't' is given by $y = a \sin(bt - cx)$, where a, b and c are constants of the wave, which of the following is a quantity with dimensions?
(A) y/a (B) bt (C) cx (D) b/c
- If $y = x \ln x$ then $\frac{dy}{dx}$ will be
(A) $\ln x + x$ (B) $1 + \ln x$ (C) $\ln x$ (D) 1
- Differentiation of $\sin(x^2 + 3)$ w.r.t. x is
(A) $\cos(x^2 + 3)$ (B) $2x \cos(x^2 + 3)$
(C) $(x^2 + 3) \cos(x^2 + 3)$ (D) $2x \cos(2x + 3)$
- Equation of straight line is $2x + 3y = 5$. Slope of the straight line is
(A) $3/2$ (B) $2/3$ (C) $-2/3$ (D) $-3/2$
- The expression ML^2T^{-2} represents :
(A) Power (B) Kinetic energy (C) Momentum (D) Pressure
- Find the maximum value of function $y = 9 - (x - 3)^2$
(A) 9 (B) 8 (C) 3 (D) 6
- Find the minimum value of $y = x + \frac{1}{x}$ for $x > 0$
(A) 2 (B) -2 (C) 1 (D) -1
- If $y = \sin 2\theta$. Find $\frac{dy}{d\theta}$
(A) $\cos 2\theta$ (B) $2 \cos 2\theta$
(C) $2 \sin 2\theta$ (D) $\tan 2\theta$
- Find derivate of $y = x^2 \sin x$
(A) $x(2\sin x + x \cos x)$ (B) $(\sin x + x \cos x)$
(C) $(2\sin x + x \cos x)$ (D) $x(2\cos x + x \sin x)$
- If $y = x^3 \tan x$ find $\frac{dy}{dx}$
(A) $x^3 \sec x + 3x^2 \tan x$ (B) $x^3 \sec^2 x + 3x^2 \tan x$
(C) $x^2 \sec x + 3x \tan x$ (D) $x \sec x + 3 \tan x$
- $y = x(2 - x)$, Find maximum value of y.
- For the equation $F = A^a v^b d^c$ where F is force, A is area, v is velocity and d is density, with the help of dimensional analysis calculate the value of $(abc) = ?$
- The frequency f of vibrations of a mass m suspended from a spring of spring constant K is given by $f = C m^x K^y$, where C is a dimensionless constant. The values of $(x + y)$ is
- If $y = t^3$ then $\frac{d^2y}{dt^2}$, If t = 1 is 2n then the value of n is
- Dimension analysis of equation $(\text{velocity})^x = (\text{pressure difference})^{3/2} (\text{density})^{-3/2}$ gives the value of x is

Rough Work

CHEMISTRY**Max. Marks: 60**

16. Hemoglobin contains 0.34% of iron by mass. The number of Fe atoms in 3.5 gm of hemoglobin is
(Given: At. Mass of Fe atom is 56 u & $N_A = 6.022 \times 10^{23}$ atoms per mole.)
(A) 1.21×10^5 (B) 12.0×10^{16} (C) 1.27×10^{20} (D) 3.4×10^{22}
17. The % of copper and oxygen in samples of CuO obtained by different methods were found to be the same. This proves the law of
(A) constant proportion (B) multiple proportion
(C) Reciprocal proportion (D) conservation of mass
18. From 320 mg of O_2 , 6.023×10^{20} molecules are removed, the number of moles remaining are
(A) 9×10^{-3} moles (B) 9×10^{-2} moles (C) zero (D) 3×10^{-3} moles
19. Perform the following calculation to proper number of significant digits
 $0.0200 \text{ km} + 4.2 \times 10^2 \text{ cm}$
(A) 24.2 m (B) 24 m (C) 25 m (D) 24.200 m
20. What will be total mass of 'N' atoms in 0.6 moles of $K_4[Fe(CN)_6]$?
(A) 5.04 gm (B) 50.4 gm (C) 10.8 gm (D) 100.8 gm
21. Element A & B form two compounds B_2A_3 and B_2A . Given that 0.05 moles of B_2A_3 weigh 9.0 gm and 0.1 mole of B_2A weigh 10 gm. Atomic weight of A & B are
(A) 20 and 30 (B) 30 and 40 (C) 40 and 30 (D) 30 and 20
22. 50 gm of $CaCO_3$ sample decomposes on heating to give CaO and 11 gram CO_2 . What will be the percentage purity of $CaCO_3$ in the sample?
(A) 60 (B) 40 (C) 55 (D) 50
23. Ratio of moles of methanol and acetic acid containing equal moles of oxygen atoms will be
(A) 2 : 1 (B) 1 : 1 (C) 1 : 2 (D) 3 : 1
24. Number of moles of $C_2H_4O_2$ containing 6.02×10^{23} atoms of hydrogen.
(A) 0.5 moles (B) 0.25 moles (C) 1 mole (D) 4 mole
25. Which of the following have same number of significant figures?
(I) 0.00253 (II) 1.003 (III) 15.0 (IV) 163
Choose the correct answer from the options given below:
(A) I, II & III (B) III & IV (C) I, III & IV (D) II & III
26. 3.011×10^{22} atoms of an element weight 1.15 gm. The atomic mass of the element is ____ amu.
27. 5 L of an alkane requires 25 L of oxygen for its complete combustion. If all volume are measured at constant temperature and pressure, find the molar mass of alkane.
28. The moles of magnesium phosphate ($Mg_3(PO_4)_2$) will contain 0.25 moles of oxygen atom is 3.125×10^{-x} moles. Find x.
29. Number of chlorine atoms in 22.4 ml of chlorine gas at STP is $x \times 10^{19}$ (Round off to the nearest integer). Find x.
30. On complete combustion, 0.492 gm of an organic compound gave 0.792 gm of CO_2 the % of carbon in the organic compound is.

MATHEMATICS**Max. Marks: 60**

31. Let α, β be the roots of $x^2 - x + p = 0$ and γ, δ be the roots of $x^2 - 4x + q = 0$. If $\alpha, \beta, \gamma, \delta$ are in GP, then the integer values of p and q respectively are
(A) -2, -32 (B) -2, 3 (C) -6, 3 (D) -6, -32
32. Let $a_1, a_2, \dots, a_{10}$ be a GP. If $\frac{a_3}{a_1} = 25$, then $\frac{a_9}{a_5}$ equals
(A) 5^3 (B) $2(5^2)$ (C) $4(5^2)$ (D) (5^4)
33. The third term of a GP is 4. The product of the first five terms is
(A) 4^3 (B) 4^5 (C) 4^4 (D) None of these
34. The ratio of sum of m and n terms of an A. P. is $m^2 : n^2$. Then the ratio of the m^{th} and n^{th} terms is
(A) $2m + 1 : 2n - 1$ (B) $m : n$ (C) $2m - 1 : 2n - 1$ (D) $2n + 1 : 2m - 1$
35. If a, b, c are in A.P. and $a + 1, b, c + 3$ are in G.P., arithmetic mean of a, b, c is 8, then the one possible value of cube of geometric mean of a, b, c is
(A) 312 (B) 314 (C) 318 (D) 128
36. The sum of n A.Ms between two given quantities is equal to n times
(A) their product (B) the square root of their product
(C) the A. M. between them (D) the G. M. between them
37. If $\frac{a^{n+1} + b^{n+1}}{a^n + b^n}$ is the G. M. between two positive numbers 'a' and 'b', then value of n is
(A) $\frac{1}{3}$ (B) $-\frac{1}{3}$ (C) $-\frac{1}{2}$ (D) $\frac{1}{2}$
38. If $x > 1, y > 1, z > 1$ are in GP, then $1 + \ln x, 1 + \ln y, 1 + \ln z$ are in
(A) AP (B) GP (C) HP (D) none of these
39. $3.9^{1/2} \cdot 27^{1/4} \cdot 81^{1/8} \dots$ is equal to
(A) 81 (B) 27 (C) 3 (D) 243
40. If $\log\left(\frac{5c}{a}\right), \log\left(\frac{3b}{5c}\right)$ and $\log\left(\frac{a}{3b}\right)$ are in AP, where a, b, c are in GP, (a, b, c > 0), then a, b, c are the lengths of sides of
(A) An isosceles triangle (B) An equilateral triangle
(C) A scalene triangle (D) None
41. If m arithmetic means (AMs) and three geometric means (GMs) are inserted between 3 and 243 such that 4th AM is equal to 2nd GM, then m is equal to
42. The minimum value of the sum of real numbers $a^{-5}, a^{-4}, 3a^{-3}, 1, a^8$ and a^{10} with $a > 0$ is ____
43. The 10th common term between the series $3 + 7 + 11 + \dots$ and $1 + 6 + 11 + \dots$ is
44. Three positive numbers form an increasing G.P. If the middle term in this G.P. is doubled, then new numbers are in A.P. The common ratio of this G.P. is $2 + \sqrt{\alpha}$, then the value of α is.
45. The lengths of three unequal edges of a rectangular solid block are in GP. The volume of the block is 216 cm^3 and the total surface area is 252 cm^2 . The length of the longest edge is ____ cm.

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